From Classroom/Integrated School Programs 7 in Top 20, 23 in Top 100, 54 in Top 300, 106 in Top 500 All India Ranks & 2314 Students from Classroom /Integrated School Programs & 3723 Students from All Programs have been Awarded a Rank in JEE (Advanced), 2013

ALL INDIA TEST SERIES

CONCEPT RECAPITULATION TEST - II

Paper 1

Time Allotted: 3 Hours **Maximum Marks: 210**

- Please read the instructions carefully. You are allotted 5 minutes specifically for this purpose.
- You are not allowed to leave the Examination Hall before the end of the test.

INSTRUCTIONS

A. General Instructions

1. Attempt ALL the questions. Answers have to be marked on the OMR sheets.
2. This question paper contains Three Parts.
3. **Part-I** is Physics, **Part-II** is Chemistry and **Part-III** is Mathematics.
4. Each part is further divided into two sections: **Section-A & Section-C**
5. Rough spaces are provided for rough work inside the question paper. No additional sheets will be provided for rough work.
6. Blank Papers, clip boards, log tables, slide rule, calculator, cellular phones, pagers and electronic devices, in any form, are not allowed.

B. Filling of OMR Sheet

1. Ensure matching of OMR sheet with the Question paper before you start marking your answers on OMR sheet.
2. On the OMR sheet, darken the appropriate bubble with black pen for each character of your Enrolment No. and write your Name, Test Centre and other details at the designated places.
3. OMR sheet contains alphabets, numerals & special characters for marking answers.

C. Marking Scheme For All Three Parts.

(i) **Section-A (01 to 10)** contains 10 multiple choice questions which have only one correct answer. Each question carries **+3 marks** for correct answer and **– 1 mark** for wrong answer.

Section-A (11 to 15) contains 5 multiple choice questions which have more than one correct answer. Each question carries **+4 marks** for correct answer. There is no negative marking.

(ii) **Section-C (01 to 05)** contains 5 Numerical based questions with answers as numerical value and each question carries **+4 marks** for correct answer. There is no negative marking.

Name of the Candidate	
Enrolment No.	

Useful Data

PHYSICS

Acceleration due to gravity	$g = 10\ m/s^2$
Planck constant	$h = 6.6 \times 10^{-34}$ J-s
Charge of electron	$e = 1.6 \times 10^{-19}$ C
Mass of electron	$m_e = 9.1 \times 10^{-31}$ kg
Permittivity of free space	$\varepsilon_0 = 8.85 \times 10^{-12}\ C^2/N\text{-}m^2$
Density of water	$\rho_{water} = 10^3\ kg/m^3$
Atmospheric pressure	$P_a = 10^5\ N/m^2$
Gas constant	$R = 8.314\ J\ K^{-1}\ mol^{-1}$

CHEMISTRY

Gas Constant	R	=	$8.314\ J\ K^{-1}\ mol^{-1}$
	=		0.0821 Lit atm $K^{-1}\ mol^{-1}$
	=		$1.987 \approx 2$ Cal $K^{-1}\ mol^{-1}$
Avogadro's Number	N_a	=	6.023×10^{23}
Planck's constant	h	=	6.625×10^{-34} J·s
	=		6.625×10^{-27} erg·s
1 Faraday		=	96500 coulomb
1 calorie		=	4.2 joule
1 amu		=	1.66×10^{-27} kg
1 eV	=		1.6×10^{-19} J

Atomic No: H=1, He = 2, Li=3, Be=4, B=5, C=6, N=7, O=8, N=9, Na=11, Mg=12, Si=14, Al=13, P=15, S=16, Cl=17, Ar=18, K =19, Ca=20, Cr=24, Mn=25, Fe=26, Co=27, Ni=28, Cu = 29, Zn=30, As=33, Br=35, Ag=47, Sn=50, I=53, Xe=54, Ba=56, Pb=82, U=92.

Atomic masses: H=1, He=4, Li=7, Be=9, B=11, C=12, N=14, O=16, F=19, Na=23, Mg=24, Al = 27, Si=28, P=31, S=32, Cl=35.5, K=39, Ca=40, Cr=52, Mn=55, Fe=56, Co=59, Ni=58.7, Cu=63.5, Zn=65.4, As=75, Br=80, Ag=108, Sn=118.7, I=127, Xe=131, Ba=137, Pb=207, U=238.

Physics

PART – I

SECTION – A

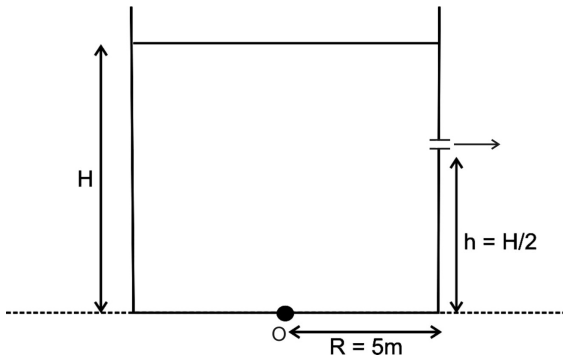
Single Correct Choice Type

This section contains **10 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE is correct**.

1. Two mutually perpendicular infinitely long straight conductors carrying uniformly distributed charges of λ_1 and λ_2 are at x = 0 along y–axis and at x = a parallel to z axis. Then:
(A) The force between them is infinite
(B) The force between them depends on 'a'
(C) The force between them is independent of 'a'
(D) The torque of force about x axis on any conductor is non zero

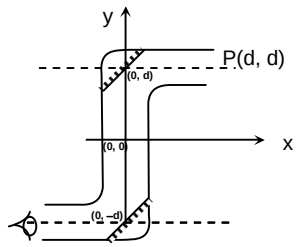
2. A block is attached to a horizontal spring of stiffness k. The other end of the spring is attached to a fixed wall. The entire system lie on a horizontal surface and the spring is in natural state. The natural length of the spring is ℓ_0. If the block is slowly lifted up vertically to a height $\frac{5}{12}\ell_0$ from its initial position, which of the following is not correct:
(A) The work done by the gravity $\frac{5}{12}\ell_0 mg$
(B) The work done by the spring force $-\frac{k\ell_0^2}{288}$
(C) The work done by the lifting force $\frac{5}{12}mg\ell_0 + \frac{k\ell_0^2}{288}$
(D) The sum of works done by all the forces on the block is zero

3. A small hole of area 1 cm^2 is punched on the side of a cylindrical vessel containing water upto a height H = 10 m as shown. The torque of normal force about 'Centre of the vessel O' immediately after making the hole would be :
Assume that tank does not topple.
(A) 10 Nm
(B) 30 Nm
(C) 50 Nm
(D) Zero



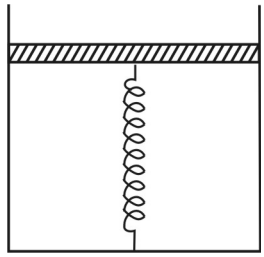
Space for rough work

4. Find the position coordinates of the image of the object P as seen through the periscope. The mirrors are inclined at 45° with x-axis and eye is placed at the level y = –d.

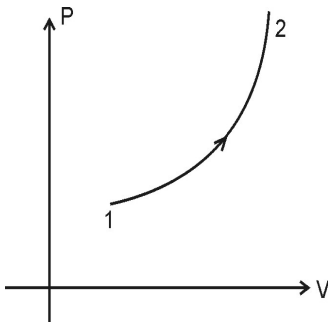


(A) (d, –d) (B) (–d, +d)
(C) (2d, –d) (D) (3d, –d)

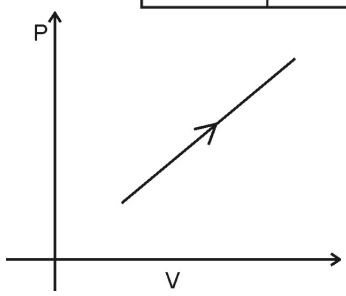
5. The piston is massless and the spring is ideal and initially streched the piston cylinder arrangement encloses an ideal gas. If the gas is heated quasistatically the PV graph.



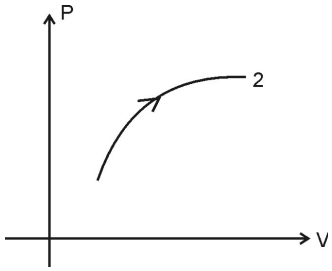
(A)



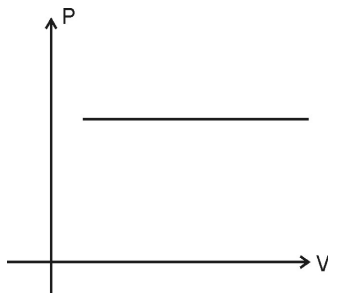
(B)



(C)



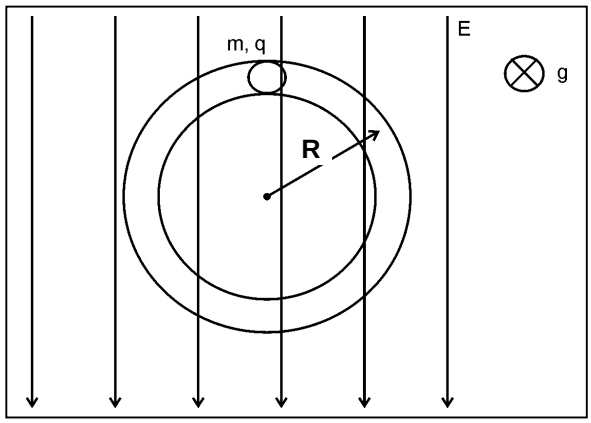
(D)



Space for rough work

6. When an excited H atom goes to ground state it emits a photon.
Statement 1 : Mass has changed into energy.
Statement 2 : K.E. has increased.
(A) Both 1 and 2 are true
(B) Both 1 and 2 are false
(C) Only 1 is true
(D) Only 2 is true

7. The smooth circular pipe is kept fixed in a horizontal plane. If mass m with change +q is slightly displaced from the position shown, then distance traveled by the ball when the magnitude of normal becomes minimum would be
(A) 0
(B) $R\cos^{-1} 1/3$
(C) $R\tan^{-1}\frac{\sqrt{5}}{2}$
(D) πR



8. When a tangential force of 0.02 N is applied on a large wooden plate of area 10 m^2 floating on the surface of river, plate moves with the speed 2 m/s on the river surface. The river is 1 m deep and the water in contact with the bed is stationary. Assuming constant speed gradient, coefficient of viscosity of water is
(A) 10^{-5} poise
(B) 10^{-2} poise
(C) 10^{-9} poise
(D) 10^{-15} poise

9. A car is moving towards an approaching bike. Bike is moving with velocity 30 m/s. Frequency of horn sounded by car is 100 Hz while it is observed by bike rider as 120 Hz. The actual duration of horn is 6 sec. The time the bike rider hears it, in seconds is (take velocity of sound in air as 330 m/s)
(A) 1 sec
(B) 10 sec
(C) 5 sec
(D) 12 sec

10. An empty plastic box of mass m is found to accelerate up at a rate of g/6 when placed deep inside water. Then the mass of sand be filled in it so that it may accelerate down at a rate of g/6 is
(A) $\frac{2}{5}m$
(B) $\frac{4}{5}m$
(C) $\frac{13}{5}m$
(D) $\frac{17}{5}m$

Space for rough work

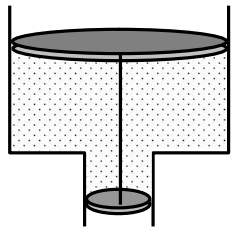
Multiple Correct Answer(s) Type

This section contains **5 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONE or MORE are correct**.

11. A block of mass 2kg moves under the influence of external forces on a rough horizontal surface. At some instant, it has a velocity of 1 m/sec. due south and acceleration of 1 m/sec^2 due east. The force of friction acting on it is F.
(A) F acts due west
(B) F acts due north
(C) F acts in the south-west direction
(D) The magnitude of F can not be found from the given data.

12. In a certain series LCR A.C. circuit it is found that $X_L = 2X_C$ and phase difference between the current and voltage is $\pi/4$. If now capacitance is made 1/4th & the inductance and resistance are doubled then which are not correct.
(A) voltage will lead current by a phase of $\tan^{-1}(1/2)$.
(B) voltage will lag behind current by a phase of $\tan^{-1}(1/2)$.
(C) voltage will lag behind current by a phase of $\pi/4$.
(D) voltage and current will be in same phase.

13. A vertical cylinder contains ideal gas with the help of two pistons whose combined mass is M. Pistons are connected by an inextensible string. The difference in the area of pistons is A. Atmospheric pressure is P_0. Initially the system is in equilibrium as shown in the figure. Now heat is supplied to the system. Mark the correct statement(s).



(A) volume will increase
(B) Pressure will remain constant
(C) Temperature will increase
(D) Internal energy of gas will remain constant.

14. As shown in the fig. all surface are frictionless and block B is released from top of the incline A then



(A) Linear momentum of (A + B) will remain constant in all direction.
(B) Linear momentum of (A + B) will remain constant in horizontal direction.
(C) Linear momentum of A will remain constant in horizontal direction.
(D) Mechanical Energy of (A + B) system will remain constant

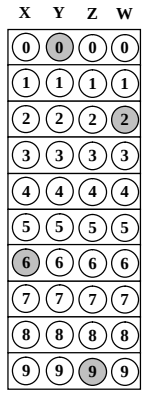
15. Choose the INCORRECT statement(s)
(A) The flux of electric field is equal to number of field lines crossing the surface
(B) The flux of electric field over a closed surface is always non-zero
(C) The flux of gravitational field over any closed surface can never be positive
(D) The flux is equal to $\phi = \vec{E}.\vec{S}$ where $\vec{E}$ is electric field and $\vec{S}$ is area of surface.

Space for rough work

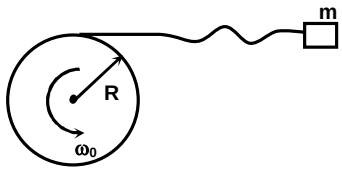
SECTION –C

Integer Answer Type

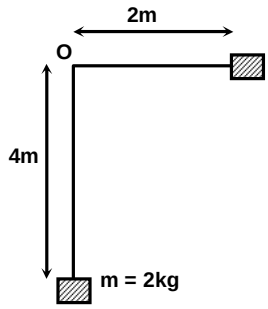
This section contains 5 questions. The answer to each of the questions is a single-digit integer, ranging from 0 to 9. The appropriate bubbles below the respective question numbers in the ORS have to be darkened. For example, if the correct answers to question numbers X, Y, Z and W (say) are 6, 0, 9 and 2, respectively, then the correct darkening of bubbles will look like the as shown.



1. Disc of mass M = 10 kg and radius R = 50 cm is revolving about fixed vertical axis on smooth horizontal surface with angular velocity ω_0 = 4 rad/sec. A block of mass m = 5kg is connected to disc through slack string as shown in the diagram. Find the velocity of the block when string become taut.



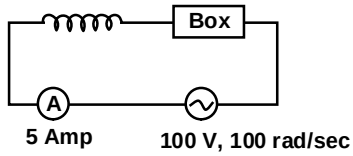
2. A block of mass 2 kg is tied to elastic thread of length 2 m and suspended to point O. Initially block is horizontal level to this point O. Now block is released and its speed is 5 m/s when thread is vertical. If the acceleration of the block at the lowest position is 2.5 K. Find the value of K.



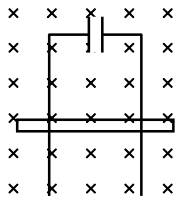
Space for rough work

3. A non conducting ring of mass m = 1 kg and radius R = 10 cm has a charge Q = 2 C uniformly distributed over its circumference. The ring is placed on rough horizontal surface such that plane of the ring is parallel to the surface. A vertical magnetic field $B = B_0t^2$ where $B_0 = 4T/sec^2$ is switched on. After 5 sec, after switching on the magnetic field the ring is just about to rotate about vertical axis through the centre. The coefficient of friction μ between the ring and the surface is found to be $x \times 10^{-1}$. Find the value of x.

4. In the circuit shown, the power factor of the circuit is 1, power factor of box is 4/5, source voltage is 100V and angular frequency ω = 100 rad/sec and current flowing in circuit is 5 amp. Value of inductance of inductor is $\frac{K}{20}$ henry then find value of K.



5. A capacitor of capacitance 100 μF is connected to smooth railing and connector of mass 1 kg and length 1 m, whole system is in vertical plane and a magnetic field of strength 100 T is applied perpendicular to plane of circuit as shown. Connector is released under gravity. The time (in sec) after which dielectric of capacitor will burn out is $k \times 10^3$ then find the value of k. Breaking strength of dielectric is 5×10^8 V/m. (d is the distance between plates of capacitor = 1 mm)



Space for rough work

Chemistry

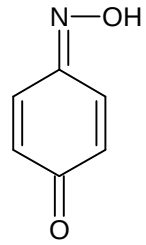
PART – II

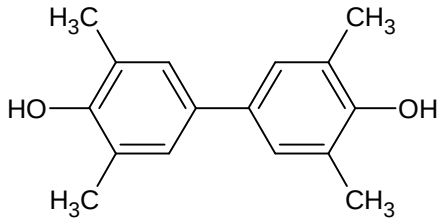
SECTION – A

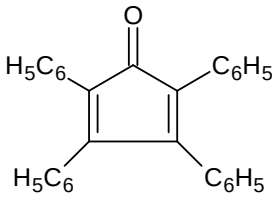
Straight Objective Type

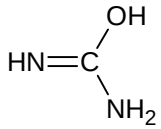
This section contains **10 multiple** choice questions numbered **1 to 10.** Each question has 4 choices (A), (B), (C) and (D), out of which **ONLY ONE** is correct.

1. Which of the following will not show tautomerism:

(A) 

(B) 

(C) 

(D) 

2. Which of the following reactions represent developing in photography?

(A) $AgNO_3 + NaBr \longrightarrow AgBr + NaNO_3$

(B) $AgBr + 2Na_2S_2O_3 \longrightarrow Na_3\left[Ag(S_2O_3)_2\right] + NaBa$

(C) $AgBr + h\upsilon \longrightarrow \dot{A}g\dot{B}r$

(D) $C_6H_4(OH)_2 + 2\dot{A}g\dot{B}r \longrightarrow C_6H_4O_2 + 2HBr + 2Ag$

3. In an adiabatic expansion of air (assume it a mixture of N_2 and O_2), the volume increases by 5% the percentage change in pressure is:

(A) 3% (B) 4%
(C) 6% (D) 7%

4. In the normal spinel having one eighth of the tetrahedral holes, occupied by one type of metal ion and one half of the octahedral holes occupied by another type of metal ion. Such a spinel is formed by Mg^{2+}, Al^{3+} and O^{2-}. If oxide ion is replaced by $X^{-8/3}$, the number of anionic vacancies per unit cell is;

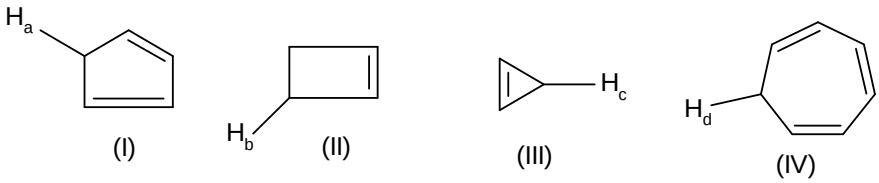
(A) 1 (B) 2
(C) 3 (D) ¾

Rough Work

5. Caffeine is dissolved in a suitable solvent and the osmotic pressure (π) of solutions of various concentrations (g/cm^3) C is measured at 20°C. The slope of a plot of π against C is found to be 4.65×10^{-3}. The molecular weight of the caffeine (g/mol) is:
(A) 3×10^5 (B) 9×10^5
(C) 4.5×10^5 (D) 5.16×10^6

6. In which of the following molecule, the number of possible $\angle$XAX angles is maximum in the anionic part of their solid state? [A : central atom, X : surrounding atom]
(A) PBr_5 (B) N_2O_5
(C) PCl_5 (D) Cl_2O_6

7. Arrange the following in increasing order of pk_o values:



(I) (II) (III) (IV)

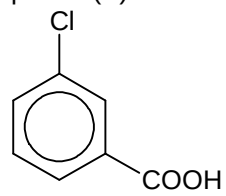
(A) (I) < (II) < (III) < (IV) (B) (I) < (II) < (IV) < (III)
(C) (IV) < (III) < (II) < (I) (D) (III) <(IV) < (II) < (I)

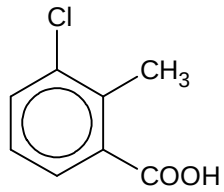
8. HCl gas is passed through an aqueous solution of 0.1 M 1 – aminopropane ($PrNH_2$) till the pH reaches 9.71. Calculate the ratio of $[PrNH_2]/[PrNH_3]^+$ in this solution:
$\left[K_a\left(PrNH_3^+\right) = 1.96 \times 10^{-11}\right]$
(A) 0.01 (B) 0.002
(C) 0.1 (D) 10^{-3}

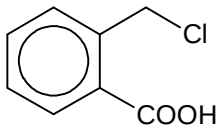
Rough Work

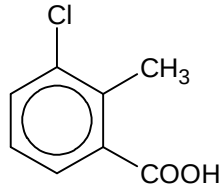
9. $\text{Compound}(A) \xrightarrow[\text{aq. NaOH}]{\text{dil}}, \xrightarrow{H^+} \xrightarrow[Na_2Cr_2O_7]{\text{acidic}} \xrightarrow{\Delta} \text{Product}(C_8H_4O_3)$

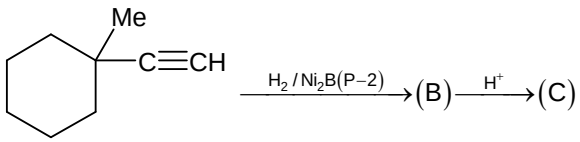
The compound (A) is:

(A) 

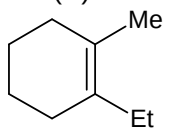
(B) 

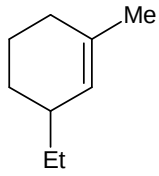
(C) 

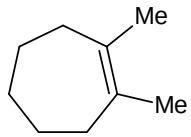
(D) 

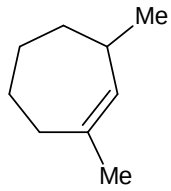
10. 

Compound (C) is:

(A) 

(B) 

(C) 

(D) 

Rough Work

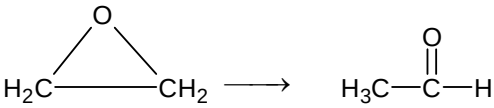
Multiple Correct Choice Type

This section contains 5 multiple choice questions. Each question has 4 choices (A), (B), (C) and (D) for its answer, out which **ONE OR MORE** is/are correct.

11. Among the isomers of dimethylcyclohexane, the achiral ones are:
(A) 1, 2-trans and 1,3 – cis (B) 1, 2-cis and 1. 3 – trans
(C) 1, 3 – trans and 1, 4 – trans (D) 1, 2 and 1, 3 – trans

12. Select the correct statement (s) about the compound $NO[BF_4]$:
(A) Magnetic moment of the species is zero
(B) B—F bond energy is lower in this compound than in BF_3
(C) It has 5σ and 2π bond
(D) Nitrogen oxygen bond order is same as in N_2

13. The oxidation potential of hydrogen half-cell will be negative if:
(A) $p(H_2) = 1$ atm and $[H^+] = 1$ M (B) $p(H_2) = 1$ atm and $[H^+] = 2$ M
(C) $p(H_2) = 0.2$ atm and $[H^+] = 1$ M (D) $p(H_2) = 0.2$ atm and $[H^+] = 0.2$ M

14. 

Given: For ethylene oxide ΔH_f^o and ΔS^o are –51 kJ mol^{-1} 243 $Jmol^{-1}K^{-1}$ respectively and acetaldehyde. ΔH_f^o and ΔS^o are –166 $KJmol^{-1}$ and 266 $Jmol^{-1}K^{-1}$ respectively, then select the correct statements:
(A) ΔH^o per the reation is –115 $KJmol^{-1}$
(B) $\Delta S^o = 0.0236$ $KJmol^{-1}K^{-1}$
(C) Reaction is thermodynamically favourable
(D) Low temperature can favour formation of more product.

15. Which of the following is/are correct about:
Tetramminedithiocyanato – S – cobalt (III) tris (oxalate) cobaltate (III)?
(A) Formula of the complex is $[Co(SCN)_2(NH_3)_4][Co(ox)_3]$
(B) It is a chelating complex and show linkage isomerism
(C) It shows optical isomerism
(D) It shows geometrical isomerism

Rough Work

SECTION – C

Integer Answer Type

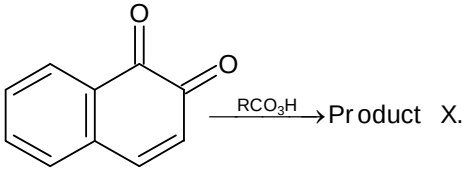
This section contains 5 questions. The answer to each of the questions is a single-digit integer, ranging from 0 to 9. The appropriate bubbles below the respective question numbers in the ORS have to be darkened. For example, if the correct answers to question numbers X, Y, Z and W (say) are 6, 0, 9 and 2, respectively, then the correct darkening of bubbles will look like the as shown.

X	Y	Z	W
0	**0**	0	0
1	1	1	1
2	2	2	**2**
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	**9**	9

1. $AgNO_3$ gives white precipitate with hypo changing to black after some time, the oxidation number of sulphur in the black precipitate is:

2. In the structure of $(Sb_2F_{11})^-$, the total number of hybrid orbitals present per Sb – atom are

3. In the given reaction:



The number of sp^2 carbon is X are expressed as a two-digit number AB, the sum of AB is:

4. The emf of the cell: $M(s)|M_nX_n(0.01M)||M_nX_n$ (0.1M)/ M_{cs}) is found to be 0.0295V at 25°C. If M_nX_n is soluble salt. Find the value of n.

5. If the bond enthalpies of A_2, AB and B_2 are in the ratio. 1 : 1 : 0.5 and the enthalpy of formation of AB from A_2 and B_2 is –100 $KJmol^{-1}$, the bond enthalpy of A_2 is 504 kJ/mole. X will be $y \times 10^2$ kJ/mole, the value of y is:

Rough Work

Mathematics

PART – III

SECTION – A

Straight Objective Type

This section contains **10 multiple choice questions** numbered 1 to 10. Each question has 4 choices (A), (B), (C) and (D), out of which **ONLY ONE** is correct.

1. Let $\vec{u}$ be a vector on rectangular coordinate system with sloping angle 60º. Suppose that $|\vec{u}-\hat{i}|$ is geometric mean of $|\vec{u}|$ and $|\vec{u}-2\hat{i}|$ where $\hat{i}$ is the unit vector along x–axis then $|\vec{u}|$ has the value equal to $\sqrt{a}-\sqrt{b}$ where a, b $\in$ N, then the value of $(a+b)^3+(a-b)^3$ is
(A) 27 (B) 28
(C) $28\sqrt{3}$ (D) $27\sqrt{3}$

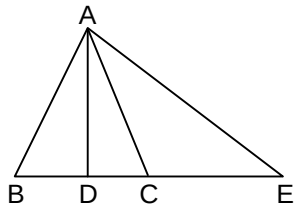
2. One of the roots of equation $2000x^6 + 100x^5 + 10x^3 + x - 2 = 0$ is of the form $\frac{m+\sqrt{n}}{r}$, where m is non–zero integer, and r and n are relatively prime natural number, then the value of (m + n + r) is
(A) 200 (B) 205
(C) 195 (D) 190

3. A perfectly rough plane is inclined at an angle α to the horizontal. Then the least eccentricity of an ellipse which can rest on the plane is
(A) $\sqrt{2\sin\alpha}$ (B) $\sqrt{\frac{2\sin\alpha}{1+\sin\alpha}}$
(C) $\sqrt{\frac{2\cos\alpha}{1+\sin\alpha}}$ (D) $\sqrt{\frac{2\sin\alpha}{1+\cos\alpha}}$

4. If the value of limit $\lim\limits_{x\to\infty}\sum\limits_{k=2}^{n}\cos^{-1}\left(\frac{1+\sqrt{k(k-1)(k+1)(k+2)}}{k(k+1)}\right)$ is equal to $\frac{120\pi}{k}$ then the value of k is
(A) 360 (B) 700
(C) 720 (D) 240

Space for rough work

5. In triangle ABC, |AB| = |AC| points D and E lie on ray BC such that |BD| = |DC| and |BE| > |CE|. Suppose that tan ∠EAC, tan ∠EAD and tan ∠EAB form a geometric progression, and that cot ∠DAE, cot ∠CAE and cot ∠DAB form an arithmetic progression. If |AE| = 10, then area of the triangle ABC is



(A) $\frac{50}{3}$ (B) 5

(C) $\frac{10}{3}$ (D) $\frac{25}{3}$

6. A circle C is tangent to the x and y–axis in first quadrant at the points P and Q respectively. BC and AD are parallel tangents to the circle with slope –1. If the points A and B are on the y–axis while C and D are on x–axis and area of figure ABCD is $900\sqrt{2}$ sq. units. The radius of the circle is

(A) $5\sqrt{2}$ (B) $6\sqrt{2}$

(C) 30 (D) 15

7. Let A be matrix $\begin{bmatrix} 1 & 1 \\ -1 & 1 \end{bmatrix}$ then the matrix $B = A^{10} - A^9 + 2A^8 - A^7 + 4A^6 - 2A^5 + 4A^4 + A^3 - A^2 + A + I$ is

(A) A (B) A + I

(C) I (D) A – I

8. Tangents are drawn to the incircle of ΔABC. Which are parallel to its sides, let p, q, r be the length of the tangents opposite to a, b, c respectively, then the value of $\frac{p}{a}+\frac{q}{b}+\frac{r}{c}$ is

(A) $\frac{1}{4}$ (B) $\frac{1}{2}$

(C) 1 (D) 2

9. A vector given by $\bar{A} = f(t)\hat{i} + g(t)\hat{j} + \hat{k}$ moves such that it is always parallel to the vector $\vec{B} = -f''(t)\hat{i} + f'(t)\hat{j} + \hat{k}$ then magnitude of $\vec{A}$ is

(A) constant (B) a quadratic function of time

(C) a linear function of time (D) a cubic function of time

10. Let $A = \begin{vmatrix} a & b & c \\ b & c & a \\ c & a & b \end{vmatrix}$ such that |A| = 0. If a, b, c are distinct, then the sum of the coordinates of the fixed point through which the line ax + by + c = 0 passes is

(A) 2 (B) 3

(C) 4 (D) 5

Space for rough work

Multiple Correct Choice Type

This section contains **5 multiple choice questions.** Each question has 4 choices (A), (B), (C) and (D) for its answer, out which **ONE OR MORE** is/are correct.

11. If a continuous function f(x) satisfies the relation $\int_0^x t\, f(x-t)dt = \int_0^x t\, g(t)dt + \sin x + \cos x - x - 1$ for all x then which of the following does not hold good?

(A) f(0) = 1 (B) f′(0) = 0

(C) f″(0) = 2 (D) $\int_0^x f(x)dx = e^{\pi}$

12. From an urn containing x white and y–x black balls, n balls have been lost, then probability of drawing a white ball

(A) before the loss is $\frac{x}{y}$ (B) after the loss is $\frac{x}{y}$ for n = 1

(C) after the loss is $\frac{x}{y}$ for n = 2 (D) after the loss is $\frac{x}{y}$ for n = 3

13. Let u, v, w, z be complex number such that |u| < 1, |v| = 1 and $w = \frac{v(u-z)}{\bar{u}z-1}$ then

(A) $|w| \le 1 \Rightarrow |z| \le 1$ (B) $|w| \ge 1 \Rightarrow |z| \ge 1$

(C) $|w| \le 1 \Rightarrow |z| \ge 1$ (D) $|w| \ge 1 \Rightarrow |z| \le 1$

14. The function defined as $f(x) = \lim_{n\to\infty} \begin{cases} \cos^{2n} x & \text{If } x < 0 \\ \sqrt[n]{1+x^n} & 0 \le x \le 1 \\ \frac{1}{1+x^n} & x > 1 \end{cases}$ which of the following does not hold good

(A) continuous at x = 0, but discontinuous at x = 1
(B) continuous at x = 1, but discontinuous at x = 0
(C) continuous at both x = 1 and x = 0
(D) discontinuous at both x = 1 and x = 0

15. Let 2sin x + 3 cos y = 3, and 3 sin y + 2 cos x = 4 then

(A) $x + y = (4n+1)\frac{\pi}{2}$, $n \in I$

(B) $x + y = (2n+1)\frac{\pi}{2}$, $n \in I$

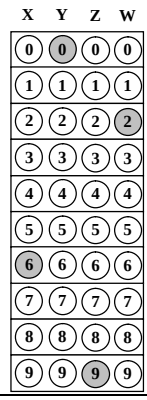
(C) x and y can be two non right angle of a 3–4–5 triangle with x > y
(D) x and y can be two non right angles of a 3–4–5 triangle with y > x

Space for rough work

SECTION – C

Integer Answer Type

This section contains **5 questions.** The answer to each of the questions is a single-digit integer, ranging from 0 to 9. The appropriate bubbles below the respective question numbers in the ORS have to be darkened. For example, if the correct answers to question numbers X, Y, Z and W (say) are 6, 0, 9 and 2, respectively, then the correct darkening of bubbles will look like the as shown.



1. The value of $\lim_{n\to\infty}\sum_{r=n^2}^{(n+1)^2}\frac{1}{\sqrt{r}}$ is _____

2. If r_1, r_2 and r_3 are three values of r which satisfy the equation $\int_0^{\pi/2}(\sin x + r\cos x)^3\,dx - \frac{4r}{\pi-2}\int_0^{\pi/2} x\cos x\,dx = 2$ then the value of $\frac{4}{7}\left(r_1^2+r_2^2+r_3^2\right)$ is equal to _____

3. Die A has four red faces and two white faces, die B has two red and four white faces. A fair coin is tossed. If it shows head, then die A is rolled thrice, otherwise die B is rolled thrice. If the first two throws of die shows red face, then the probability of getting red face in third throw is $\frac{a}{b}$ (in lowest terms) then $(a-1)^{b-3}$ is equal to _____

4. The set of all points where f(x) is increasing in $(a, b) \cup (c, \infty)$ then [a + b + c] is (where [.] denotes the greatest integer function). Given that $f(x) = 2f\left(\frac{x^2}{2}\right) + f\left(6-x^2\right)\ \forall\ x \in R$ and $f''(x) > 0\ \forall\ x \in R$ _____

5. A circle of radius R is tangent to the hypotenuse, the incircle and one leg of an isosceles right angle triangle with inradius $r = 1+\sin\frac{\pi}{8}$. If the value of $Rr = \frac{a+\sqrt{a}}{b}$, where $\sqrt{a}$ is surd then (a + b) is equal to _____

Space for rough work